

Cell-level audit records expose agreement, conflict and bounded local action

(a) Agreement record

Donor P8 | seed 42
E2L6_GTCACGGTCCTTGAAG

True **Eryth**

Final **Eryth**

Branches
RNA=Eryth; ADT=Eryth; fusion=Eryth

uncertainty=0.00003 | conflict=0
HSR action: none recorded

correct agreement; review record, not a guarantee

(b) Modality conflict

Donor P4 | seed 41
L4_TTGTGGACACGCGTCA

True **Treg Naive**

Final **CD4 Naive**

Branches
RNA=CD4 TCM_1; ADT=Treg Naive; fusion=CD4 Naive

uncertainty=0.640 | conflict=0.667
HSR group: not applicable

conflict flag; final output is wrong

(c) Bounded HSR action

Donor P1 | seed 41
L1_CAATGACAGTCTCTGA

True **CD4 TCM_1**

Final **CD4 TCM_3**

Branches
RNA=not recorded; ADT=not recorded;
fusion=not recorded

base -> final: CD4 TCM_1 -> CD4 TCM_3
gate=0.0204 | ||delta||2=12.940

sparse local action; final output is wrong